

Extending the Spatio-Temporal Skew- t Process for Threshold Exceedance Prediction

AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions

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Outline

- ① Motivation
- ② Model
- ③ Simulation
- ④ Ozone data
- ⑤ Conclusions

Problem background

- Monitoring networks: limited coverage $\rightarrow$ statistical extrapolation, with quantified confidence
- Monitoring and compliance
- Target: $\Pr(Y > L)$ at **unmonitored** sites — the upper tail, not the mean field

Ozone NAAQS (ppb)	1997	2008	2015 (current)
Annual 4th-highest daily max 8-hr avg., 3-year mean	80	75	70

Why standard spatial methods fall short

- ① **Margin** — Gaussian: light-tailed, symmetric; ozone maxima: **heavy-tailed, right-skewed**
- ② **Extremal Dependence** (Coles, Heffernan, and Tawn, 1999):

$$\chi(h) = \lim_{c \rightarrow c^*} \Pr[Y(\mathbf{s} + h) > c \mid Y(\mathbf{s}) > c], \quad \text{Gaussian: } \chi(h) = 0 \quad \forall h$$

heavy tails + asymmetry + asymptotic dependence $\Rightarrow$ **skew- t process**

Why not max-stable or EVT threshold models

- Max-stable likelihood: infeasible at 1,089 stations
- 75 ppb $\approx$ 0.92 sample quantile — not the far tail in EVT sense
- Subasymptotic setting (Huser and Wadsworth, 2022): joint tail at high but *finite* levels
- Skew- t = location-scale mixture of GPs $\in$ this family

Baseline STP: model and parameters

Morris et al. (2017):

$$Y_t(\mathbf{s}) = \underbrace{\mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta}}_{\text{regression mean}} + \underbrace{\lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})|}_{\text{skewness term}} + \underbrace{\sigma_t(\mathbf{s}) v_t(\mathbf{s})}_{\text{spatial process}}$$

- $\lambda \in \mathcal{R}$: skewness · $z_t(\mathbf{s}) \sim N(0, 1)$, $|z_t|$ half-normal
- $\sigma_t^2(\mathbf{s}) \sim \text{IG}(a/2, b/2)$: scale mixing $\rightarrow$ heavy tails ($a = \text{df}$)
- $v_t(\mathbf{s})$: standard GP, Matérn correlation (φ range, ν smoothness, η spatial proportion)
- Marginalize $|z_t|$, $\sigma_t \Rightarrow$ **skew- t margin** + asymptotic dependence

Baseline STP: Morris's extensions

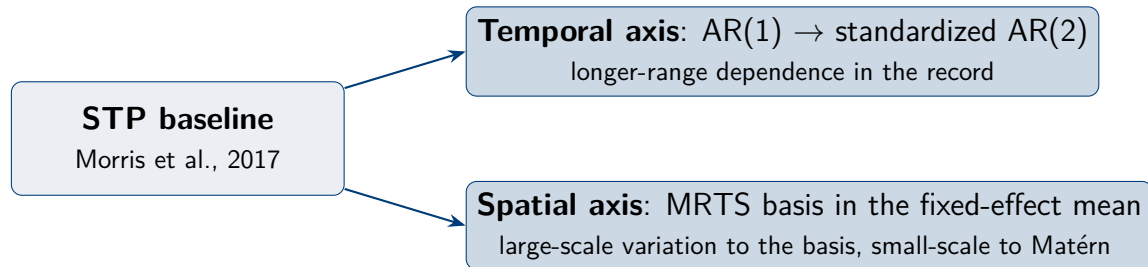
- ① **Random partition**, K knots redrawn daily:

$$P_{tk} = \{\mathbf{s} : k = \arg \min_{\ell} \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \quad z_t(\mathbf{s}) = z_{tk}, \quad \sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$$

→ **local** asymptotic dependence

- ② **Censored likelihood**: $\tilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, $T \leq L$ — upper-tail focus; below T : imputed in MCMC
- ③ **Temporal dependence**: PIT → Gaussian scale → stationary **AR(1)**; margins preserved at every t

Two candidate extensions



candidate options, not replacements — held-out scores decide

Extension 1: a standardized AR(2) prior

On the PIT-transformed Gaussian scale:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \quad (\phi_1, \phi_2) \in \text{stationarity triangle}$$

Inverse PIT needs $\gamma_0 = \text{Var}(Y_t) = 1 \Rightarrow \sigma_\epsilon^2$ **not free** (Yule–Walker):

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad \sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2$$

- Initial pair (Y_1, Y_2) : stationary bivariate Gaussian, correlation γ_1
- Identical margins across i.i.d. / AR(1) / AR(2)

Extension 2: MRTS basis functions in the mean

MRTS (Tzeng and Huang, 2018):

- Ordered **coarse** $\rightarrow$ **fine**: $f_1 = 1$; $f_2, f_3 =$ coordinates; $k \geq 4$: eigenvectors of the TPS kernel, eigenvalues decreasing
- Closed form at *new* locations $\rightarrow$ predictive covariate

$$\mathbf{X}_t^*(\mathbf{s}) = [\mathbf{f}(\mathbf{s})^\top, \mathbf{X}_t(\mathbf{s})^\top]^\top, \quad Y_t(\mathbf{s}) = \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s})$$

- Claim: non-stationary **mean** only — Matérn keeps small-scale dependence

Hierarchical model and inference

Three stages (Berliner, 1996):

Stage 1 censored data: $\tilde{Y}_t(\mathbf{s}_i) = \max\{Y_t(\mathbf{s}_i), T\}$

Stage 2 Gaussian field · partition · latent AR(2) dynamics

Stage 3 priors; $(\phi_1, \phi_2) \sim \text{Unif}(\text{stationarity triangle})$

$$[\mathbf{Y}_{\text{cens}}, \mathcal{Z}, \boldsymbol{\theta} \mid \tilde{\mathbf{Y}}] \propto \prod_{t=1}^{n_t} [\tilde{\mathbf{Y}}_t \mid \mathbf{Y}_t] [\mathbf{Y}_t \mid \mathcal{Z}_t, \boldsymbol{\theta}] [\mathcal{Z}_{1:n_t} \mid \boldsymbol{\theta}] [\boldsymbol{\theta}]$$

- Gibbs-within-MH; AR(2) full conditionals: appendix
- K fixed = **candidate model index**, chosen by held-out Brier

Simulation design

144 sites on $[0, 10]^2$ · 50 days · 10 data sets · 100 train / 44 test

Analysis models	Gaussian; skew- t ($K=1, 5$); sym- t , $T=q(0.80)$ ($K=1, 5$)
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AR(2) designs	$(0.80, -0.35)$, $(0.12, -0.05)$ at $K = 1, 5$; persistent $(0.15, 0.80)$; ARFIMA $d = 0.45$
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MRTS designs	2× second-moment non-stationarity; 2× nonlinear mean surface
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Evaluation protocols

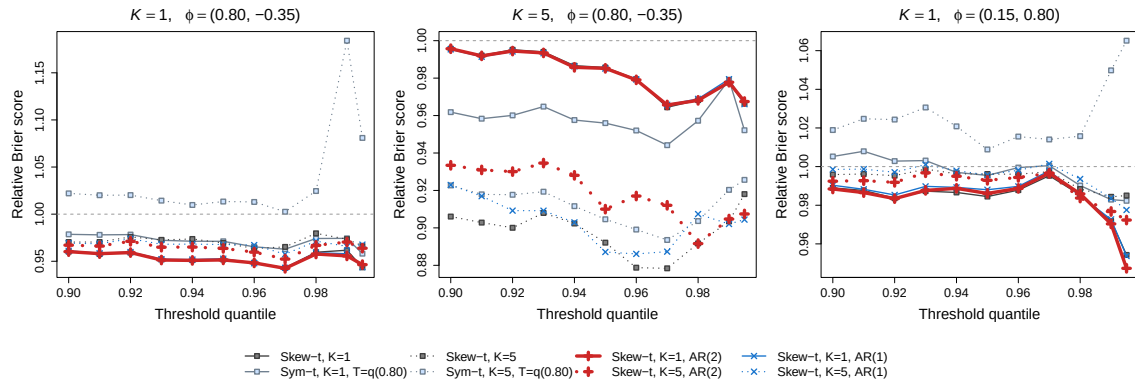
Scoring (Gneiting and Raftery, 2007):

$$\text{BS}(L) = \{I(y > L) - \hat{P}(Y > L)\}^2 \quad (\text{lower is better})$$

Protocol	Observed	Predicted	Task
Spatial hold-out	training sites, full record	unseen sites	interpolation
Time-block forecast	all sites, days 1–50	same sites, next 15 days	extrapolation

- Uncertainty: **paired** differences · 95% CI · Wilcoxon

AR(2), interpolation: temporal structure barely moves the scores



32/33 paired intervals cover zero

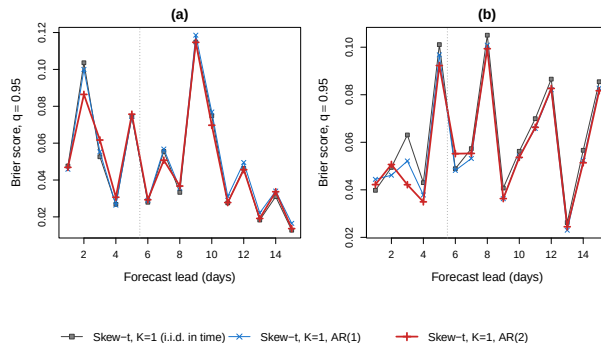
(all six settings: backup)

AR(2): coefficients identified, scores still unmoved

Generating design	$\hat{\phi}^{(\sigma)}$		$\hat{\phi}^{(z)}$	
	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$
Persistent (0.15, 0.80)	0.112 (0.132)	0.742 (0.117)	0.175 (0.152)	0.728 (0.112)
ARFIMA, $d = 0.45$	0.502 (0.237)	0.200 (0.239)	0.365 (0.249)	0.156 (0.169)

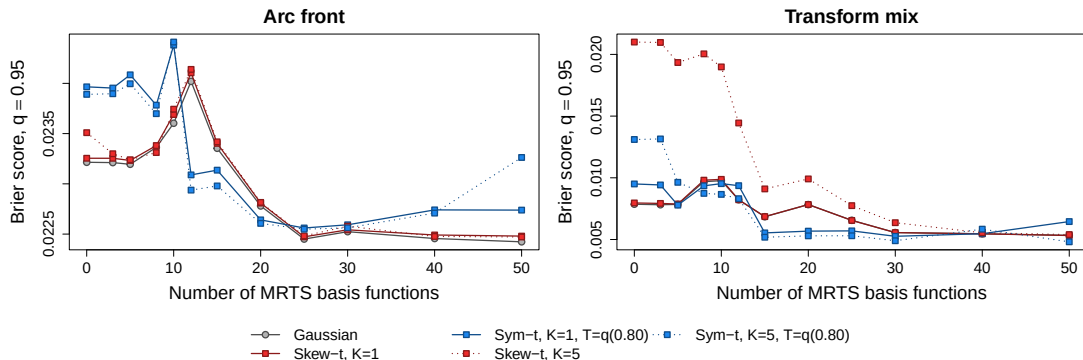
- Second lag **recovered** — even under long memory
- Hold-out scores: still unmoved $\rightarrow$ the **task**, not the model

AR(2), extrapolation: directional gains where predicted



- Long memory: AR(2) lowest at almost every lead; short-memory control: ties
- All differences n.s. $\rightarrow$ **directional** (largest -0.0068 , CI $[-0.033, 0.020]$)

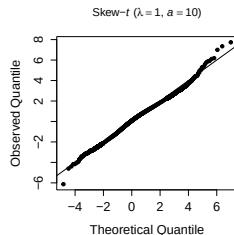
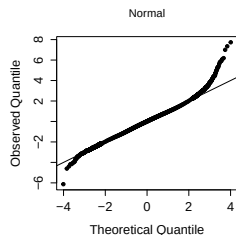
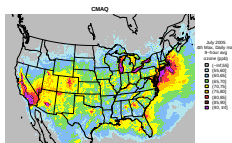
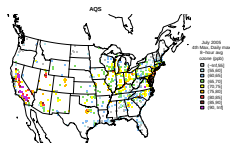
MRTS: inert against second moments, paid off once resolved



- Second-moment designs: flat — **no first moment to recover** (backup)
- Nonlinear mean: pays only once the basis **resolves** the surface; mid-size dip

Ozone data and motivating diagnostics

- July 2005 · AQS daily max 8-h ozone · **CMAQ** covariate
- Models **calibrate CMAQ** at the monitors



- Residuals: heavy-tailed, right-skewed
- Skew- t reference ($\lambda = 1$, $a = 10$): straightens

Candidate set and validation design

- Baseline grid (Morris et al., 2017): $\{\text{Gaussian, skew-}t\} \times K \in \{1, \dots, 15\} \times T \in \{0, 50, 75\} \text{ ppb} \times \{\text{TS, no TS}\}$
- Extensions: AR(2) prior; MRTS with $K_{\text{MRTS}} = 10$ **pre-specified**
- $1,089 \rightarrow 800$ **region-stratified** sites · two-fold CV (400/400) · splits 300/100, 200/200
- MCMC: 30,000 iterations, 25,000 burn-in

Results: scattered wins, but systematic

Level L	Rank 1 model	Rank 2 model
0.900	MRTS skew- t , $K=1$, $T=0$, AR(1), $K_{\text{MRTS}}=10$	skew- t , $K=1$, $T=0$, AR(1)
0.950	AR2 skew- t , $K=7$, $T=50$	AR2 skew- t , $K=10$, $T=50$
0.980	skew- t , $K=6$, $T=50$, AR(1)	skew- t , $K=5$, $T=50$, AR(1)
0.990	symmetric- t , $K=9$, $T=75$, AR(1)	skew- t , $K=5$, $T=50$
0.995	AR2 symmetric- t , $K=15$, $T=75$	symmetric- t , $K=9$, $T=75$, AR(1)

- Deeper tail $\rightarrow$ higher censoring, larger K
- Pattern reproduces on the 300/100 and 200/200 splits

Uncertainty: the two axes part ways

Site-clustered bootstrap ($B = 2000$, by fold), 400/400 split, $\Delta \times 10^3$ [95% CI]:

L quantile	AR(2) – AR(1)	MRTS – none
0.900	–0.15 [–0.38, +0.08]	–0.03 [–0.13, +0.07]
0.950	–0.12 [–0.29, +0.04]	– 0.08 * [–0.14, –0.02]
0.980	+0.03 [–0.08, +0.14]	–0.04 [–0.07, +0.00]
0.990	–0.06 [–0.14, +0.03]	– 0.03 * [–0.05, –0.00]
0.995	–0.03 [–0.08, +0.02]	–0.01 [–0.02, +0.01]

- AR(2): every interval covers zero (all splits)
- MRTS: negative everywhere; stars survive **block resampling**; $\approx 3\text{--}4 \times 10^{-3}$ of baseline

Conclusions

- **Temporal axis:** divides along the task — identified yet inert in interpolation; gains only in **forecasting**, never significant
- **Spatial axis:** **real, though modest** — needs mean-level structure, and a basis fine enough to resolve it
- **Contribution:** STP baseline → **candidate model class**, selected by held-out threshold prediction

Limitations and future work

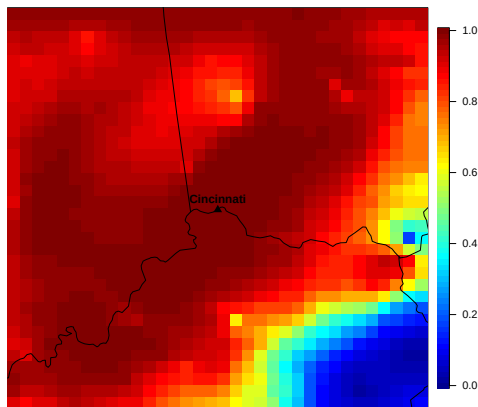
Limitations

- One month, one pollutant · single forecast origin

Future work

- **Random-coefficient** MRTS (fixed-rank kriging) → the second-moment layer
- Reparameterized z , longer records , rolling forecast origin

An exceedance-probability map



Pr(≥ 1 day > 75 ppb), July 2005, Cincinnati — skew- t + TS + MRTS ($K_{\text{MRTS}}=10$)

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Thank You

Questions & Comments Welcome

Backup

Backup: λ - z reflection ridge and forecast screening

- A reflection-type non-identifiability links λ and z : short-record fits can land on the **reflected ridge**, where the skewness parameter and the intercept fail to separate and the predictive spread inflates.
- The screening rule is public and stated with the forecast protocol: fits whose λ and intercept fail to separate, or whose predictive spread is unbounded, are removed *before* scoring.
- Root fix — a sign-anchored reparameterization of z — is listed as future work.

Backup: AR(2) stationary moments and full conditionals

- Yule–Walker under $\gamma_0 = 1$: $\gamma_h = \phi_1\gamma_{h-1} + \phi_2\gamma_{h-2}$ for $h \geq 1$ gives $\gamma_1 = \phi_1/(1 - \phi_2)$, $\gamma_2 = \phi_1\gamma_1 + \phi_2$, and $\sigma_\epsilon^2 = 1 - \phi_1\gamma_1 - \phi_2\gamma_2$.
- Gibbs-within-MH: the full conditional of Y_t carries **three** prior factors,

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} \mid Y_t) \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}};$$

omitting the lag-2 factor distorts the stationary distribution.

- Head of the series ($t = 1, 2$): the stationary initial-pair density replaces the self factor. Proposals: symmetric Gaussian random walk tuned to 0.3–0.4 acceptance; (ϕ_1, ϕ_2) updated jointly under the uniform prior on the stationarity triangle.

Backup: second-moment design — “inert” is not “no signal”

- The random effect superposes two fixed cosine fields with daily-redrawn Gaussian weights:

$$u_t(\mathbf{s}) = \sum_{j=1}^2 \xi_{tj} g_j(\mathbf{s}), \quad \xi_{tj} \stackrel{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3),$$

giving the location-dependent covariance $\text{Cov}\{u_t(\mathbf{s}), u_t(\mathbf{s}')\} = \sum_j \tau_j^2 g_j(\mathbf{s}) g_j(\mathbf{s}')$.

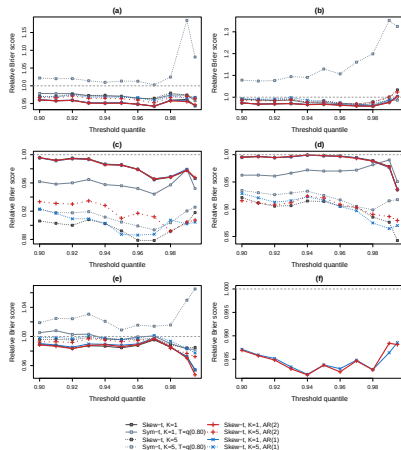
- Zero time-average $\Rightarrow$ the pooled coefficient has **no first-moment structure to recover**, while u_t still contributes substantial non-stationary variance: structurally out of reach for a fixed-effect mean, not absent signal.
- The two nonlinear-mean surfaces were confirmed by an **oracle calculation before any fitting** to leave genuine Brier headroom.

Backup: censoring pairing and MCMC settings

- Why symmetric- t with censoring: once the lower tail is censored, the skewness parameter is only **weakly identified**, so censored settings pair with a symmetric- t marginal, following Morris et al. (2017).
- Thresholds: the censor level $T = 75$ ppb sits below the era standard $L = 80$ ppb ($T \leq L$), and every evaluation level lies above T ; the degrees of freedom a control the tail thickness.
- Burn-in 25,000 of 30,000 ($\approx 83\%$): mixing is slowed by censoring imputation and the latent volume; traces were inspected, and the long burn-in is a conservative choice.

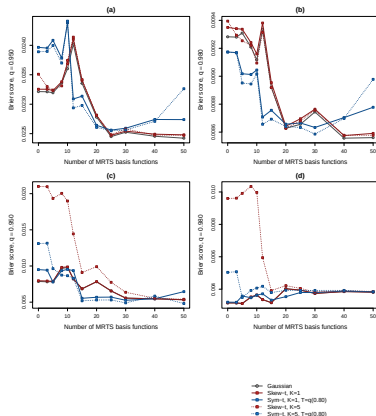
Backup: spatial hold-out, all six generating settings

Relative Brier score vs. the Gaussian reference ($\equiv 1$, dashed).



- (a)–(d): fixed AR(2) designs, $K \in \{1, 5\}$, $(\phi_1, \phi_2) = (0.80, -0.35)$ or $(0.12, -0.05)$.
- (e): persistent (0.15, 0.80).
- (f): ARFIMA, $d = 0.45$ (only the $K=1$ AR(1)/AR(2) fits).

Backup: MRTS nonlinear-mean designs, full grid



Brier score vs. K_{MRTS} ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline).

- (a, b): arc front, $q = 0.95 / 0.98$.
- (c, d): transform mix, $q = 0.95 / 0.98$.
- At $q = 0.98$ the gain is carried by the five-knot models — the sharpness–extrapolation signature.